

Session 08 (AIML) - Assignment Questions

Q1. (Creating Sets)

Write a program to create the following sets:

- a) A set with 5 integers.
- b) A set with mixed data types (integer, string, float).
- c) An empty set using the correct method.
- d) A set from the string "hello" using the set() constructor.

Print each set and its type. Add comments explaining why sets automatically remove duplicates.

Q2. (Characteristics of Sets)

Create a set: `numbers = {10, 20, 30, 20, 40, 10, 50}`

- Print the set and observe the output.
 - Demonstrate that sets are unordered by printing the set multiple times.
 - Try to access the first element using indexing (`numbers[0]`) and explain the error in comments.
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Q3. (Membership Testing)

Write a program that:

- Takes a set of 5 colors as input from the user.
 - Asks the user to enter a color name.
 - Checks whether the color is present in the set using `in` and `not in`.
 - Prints appropriate messages.
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Q4. (add(), remove(), discard())

Create a set: `fruits = {'apple', 'banana', 'mango'}`

Perform the following operations and print the set after each step:

- a) Add 'orange' using `add()`.
- b) Add 'banana' again (observe the result).

- c) Remove 'mango' using `remove()`.
 - d) Try to remove 'grape' using `discard()` (no error should occur).
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Q5. (`pop()` and `clear()`)

Create a set: `s = {100, 200, 300, 400, 500}`

- Use `pop()` to remove and print one element.
- Print the set after popping.
- Clear the entire set using `clear()`.
- Print the final empty set.

Add comments explaining the difference between `remove()`, `discard()`, and `pop()`.

Q6. (`update()` and `copy()`)

Create two sets:

`set1 = {1, 2, 3}`

`set2 = {3, 4, 5, 6}`

- Make a copy of `set1`.
 - Update `set1` with elements from `set2` using `update()`.
 - Print both the original copy and the updated set.
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Q7. (Union and Intersection)

Create two sets:

`A = {1, 2, 3, 4, 5}`

`B = {4, 5, 6, 7, 8}`

- Print the **union** of A and B (using both `.union()` and `|` operator).
- Print the **intersection** of A and B (using both `.intersection()` and `&` operator).

Add comments explaining what each operation does.

Q8. (Combined Methods)

Write a program that:

- Takes 6 numbers as input from the user and stores them in a set (duplicates should be removed automatically).

- Adds two more numbers using `add()`.
 - Removes one number safely using `discard()`.
 - Prints the final set and its length using `len()`.
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Q9. (Practical Application)

Write a program to remove duplicate marks from a list of student scores using a set:

```
scores = [85, 92, 78, 92, 85, 88, 95, 78]
```

- Convert the list to a set.
 - Print the unique scores.
 - Convert the set back to a sorted list and print it.
 - Print the total number of unique scores.
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Q10. (Mini Project - Combined Concepts)

Create a **Unique Item Collector** program using sets:

- Repeatedly ask the user to enter items (e.g., favorite fruits, colors, or numbers).
- Store all entered items in a set (duplicates should be ignored).
- Provide a menu with options:
 1. Add item
 2. Remove item (use `discard`)
 3. Show all unique items
 4. Check if an item exists
 5. Clear all items
 6. Exit

Use a while loop for the menu and appropriate set methods. Add meaningful comments.